I Limits of sequences

Definition

A sequence (u_n) is **convergent** to L if $\lim_{n\to\infty} u_n = L$. It is **divergent** if this limit does not exist or equals $\pm\infty$.

II Limits of functions

If $f(x)$ can be made as close as we like to some real number A by making x sufficiently close to (but not equal to) a, then we say that $f(x)$ has a **limit** of A as x approaches a, and we write

$$\lim_{x\to a} f(x) = A.$$

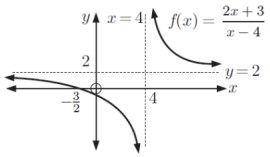
In this case, $f(x)$ is said to *converge* to A as x approaches a.

Consider the function

$$f(x) = \frac{2x+3}{x-4},$$

which has domain $\{x \mid x \neq 4,\ x \in \mathbb{R}\}$.

From the graph of $y = f(x)$ we can see that the function has a vertical asymptote $x = 4$ and a horizontal asymptote $y = 2$.



Both asymptotes can be described in terms of limits:

$$\text{As } x \to 4^-, \quad f(x) \to -\infty$$
$$\text{As } x \to 4^+, \quad f(x) \to +\infty$$
$$\text{As } x \to -\infty, \quad f(x) \to 2^-$$
$$\text{As } x \to +\infty, \quad f(x) \to 2^+$$

Since $f(x)$ converges to a finite value as $x \to \pm\infty$, we write $\lim_{x\to-\infty} f(x) = 2$ and $\lim_{x\to+\infty} f(x) = 2$.

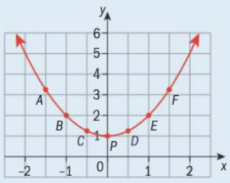
Definition

A **vertical asymptote** is a line $x = a$ such that $\lim_{x\to a^{\pm}} f(x) = \pm\infty$.
A **horizontal asymptote** is a line $y = L$ such that $\lim_{x\to\pm\infty} f(x) = L$.

III Derivative function

Investigation – secant and tangent lines

Here is the graph of $f(x) = x^2 + 1$



1 Copy the graph to paper and draw lines *AP*, *BP*, *CP*, *DP*, *EP* and *FP*. These lines are called **secant lines** to the graph of $f(x) = x^2 + 1$.

2 Copy and complete the table.

Point	Coordinates	Line	Gradient or slope
P		—	—
A		*AP*	
B		*BP*	
C		*CP*	
D		*DP*	
E		*EP*	
F		*FP*	

Recall that the gradient of a line through the points (x_1, y_1) and (x_2, y_2) is $\frac{y_2 - y_1}{x_2 - x_1}$

3 As points on the curve get closer and closer to point *P*, what value does the gradient of the secant lines seem to be approaching?

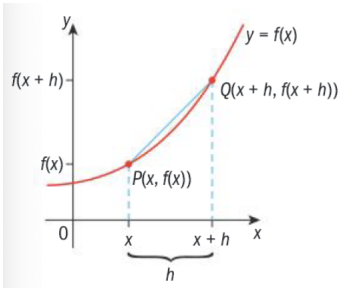
4 Draw the line at point *P* that has the gradient you found in question 3. This line is called the **tangent line** to the graph of $f(x) = x^2 + 1$ at *P*.

Definition

Given two points $P\big(x,\ f(x)\big)$ and $Q\big(x+h,\ f(x+h)\big)$ on the curve $y = f(x)$, the **gradient of the secant line** PQ is

$$\frac{f(x+h) - f(x)}{(x+h) - x} = \frac{f(x+h) - f(x)}{h}.$$

The expression $\dfrac{f(x+h) - f(x)}{h}$ is called the **difference quotient**.



Example

Write an expression for the gradient of a secant line to the curve $f(x) = x^2 + 1$ and simplify it.

Suppose that point Q slides down the curve and approaches point P. The secant lines PQ get closer to the tangent line at P. As $Q \to P$, $h \to 0$. We take the limit as $h \to 0$ of the gradient of the

secant line to obtain the gradient of the tangent line:

$$\lim_{h\to 0}\frac{f(x+h)-f(x)}{h}.$$

This limit is not a constant — it is a function that gives the gradient of f at any point x.

Definition

The function

$$f'(x)=\lim_{h\to 0}\frac{f(x+h)-f(x)}{h}$$

is the **derivative** of f.

Reading the notations. $f'(x)$ is read "f prime of x", or $\frac{dy}{dx}$ "the derivative of y with respect to x".

Remark. *When we evaluate this limit to find a derivative function, we say we are* ***differentiating from first principles****.*

Property

Gradient of the tangent at $x=a$:

$$f'(a)=\lim_{h\to 0}\frac{f(a+h)-f(a)}{h}.$$

Example

Find the derivative of $f(x)=x^2+1$ from first principles, and hence find the gradient of the tangent line when $x=3$.

IV Differentiation rules

a. Basic rules

Basic differentiation rules

For differentiable functions u, v and a constant c:

$f(x)$	Rule	$f'(x)$
c (a constant)	constant	0
x^n	power rule	nx^{n-1}
$c\,u(x)$	constant multiple	$c\,u'(x)$
$u(x)+v(x)$	sum rule	$u'(x)+v'(x)$

b. Product rule

Investigation

Suppose $u(x)$ and $v(x)$ are two functions of x and $f(x) = u(x)v(x)$. The purpose of this investigation is to find a rule for $f'(x)$.

What to do:

1. Suppose $u(x) = x$ and $v(x) = x$, so $f(x) = x^2$.
   (a) Find $f'(x)$ by direct differentiation.
   (b) Find $u'(x)$ and $v'(x)$.
   (c) Is it true that $f'(x) = u'(x)v'(x)$?
2. Suppose $u(x) = x$ and $v(x) = \sqrt{x}$, so $f(x) = x^{3/2}$.
   (a) Find $f'(x)$ by direct differentiation.
   (b) Find $u'(x)$ and $v'(x)$.
   (c) Is it true that $f'(x) = u'(x)v'(x)$?
3. Copy and complete the following table, finding $f'(x)$ by direct differentiation.

$f(x)$	$f'(x)$	$u(x)$	$v(x)$	$u'(x)$	$v'(x)$	$u'(x)v(x) + u(x)v'(x)$
x^2		x	x			
$x^{3/2}$		x	$\sqrt{x}$			
$x(x+1)$		x	$x+1$			
$(x-1)(2-x^2)$		$x-1$	$2-x^2$			

4. Copy and complete: "If $u(x)$ and $v(x)$ are two functions of x and $f(x) = u(x)v(x)$, then $f'(x) = \ldots$"

Product rule

If $u(x)$ and $v(x)$ are differentiable functions of x and $f(x) = u(x)v(x)$, then

$$f'(x) = u'(x)\,v(x) + u(x)\,v'(x).$$

c. Quotient rule

Quotient rule

If $Q(x) = \dfrac{u(x)}{v(x)}$, then

$$Q'(x) = \frac{u'(x)\,v(x) - u(x)\,v'(x)}{[v(x)]^2}.$$

Equivalently, if $y = \dfrac{u}{v}$ where u and v are functions of x:

$$\frac{dy}{dx} = \frac{u'v - uv'}{v^2}.$$

Proof

Quotient functions have the form $Q(x) = \dfrac{u(x)}{v(x)}$, so $u(x) = Q(x)\,v(x)$.

By the product rule:

$$u'(x) = Q'(x)\,v(x) + Q(x)\,v'(x).$$

$$\therefore \quad u'(x) - Q(x)\,v'(x) = Q'(x)\,v(x)$$

$$\therefore \quad Q'(x)\,v(x) = u'(x) - \frac{u(x)}{v(x)}\,v'(x) = \frac{u'(x)\,v(x) - u(x)\,v'(x)}{v(x)}$$

$$\therefore \quad Q'(x) = \frac{u'(x)\,v(x) - u(x)\,v'(x)}{[v(x)]^2}, \quad \text{when this exists.}$$

Example

Use the quotient rule to find $\dfrac{dy}{dx}$ if $y = \dfrac{1+3x}{x^2+1}$.

d. Chain rule

Chain rule

If $y = g(u)$ where $u = f(x)$, then

$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}.$$

This rule is extremely important and enables us to differentiate composite functions efficiently.

Property

For any differentiable function $f(x)$:

$$\text{If } y = [f(x)]^n \text{ then } \frac{dy}{dx} = n\,[f(x)]^{n-1} \times f'(x).$$

Example

Find $\frac{dy}{dx}$ if:

1. $y = (x^2 - 2x)^4$
2. $y = \frac{4}{\sqrt{1-2x}}$

e. Exponential functions

Theorem

If $f(x) = b^x$ then $f'(x) = k\,b^x$ where k is a constant equal to $f'(0)$.

Proof

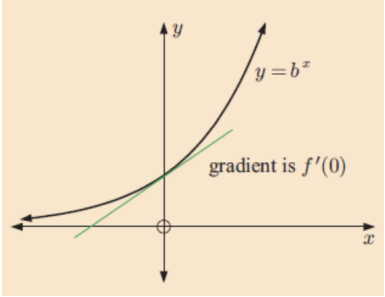
If $f(x) = b^x$, then by the first principles definition:

$$f'(x) = \lim_{h\to 0} \frac{b^{x+h} - b^x}{h} = \lim_{h\to 0} \frac{b^x(b^h - 1)}{h} = b^x \times \left(\lim_{h\to 0} \frac{b^h - 1}{h}\right).$$

Since b^x is independent of h, and since

$$f'(0) = \lim_{h\to 0} \frac{b^h - 1}{h},$$

we obtain $f'(x) = b^x\, f'(0)$.



Given this result, if we can find a value of b such that $f'(0) = 1$, we will have found a function which is its own derivative.

Investigation

Graph $f(x) = b^x$ and its derivative $y = f'(x)$.

Experiment with different values of b until the graphs of $f(x)$ and $f'(x)$ appear identical.

Estimate the corresponding value of b to 3 decimal places.

Theorem

If $f(x) = e^x$ then $f'(x) = e^x$.

f. Derivative of $e^{f(x)}$

Function	Derivative
e^x	e^x
$e^{f(x)}$	$e^{f(x)} \cdot f'(x)$

Proof

Note that $e^{f(x)} > 0$ for all x. Let $y = e^{f(x)} = e^u$ where $u = f(x)$. By the chain rule:

$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx} = e^u \cdot f'(x) = e^{f(x)} \cdot f'(x).$$

g. Derivative of $\ln f(x)$

Property

If $y = \ln x$ then $\frac{dy}{dx} = \frac{1}{x}$.

Function	Derivative
$\ln x$	$\frac{1}{x}$
$\ln f(x)$	$\frac{f'(x)}{f(x)}$

Proof

Let $y = \ln f(x) = \ln u$ where $u = f(x)$. By the chain rule:

$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx} = \frac{1}{u} \cdot f'(x) = \frac{f'(x)}{f(x)}.$$

Example

Find the gradient function of:

a) $y = \ln(kx)$, k a constant b) $y = \ln(1 - 3x)$.

Remark (Laws of Logarithms). *For $a > 0$, $b > 0$, $n \in \mathbb{R}$:*

$$\ln(ab) = \ln a + \ln b, \qquad \ln\left(\frac{a}{b}\right) = \ln a - \ln b, \qquad \ln(a^n) = n \ln a.$$

Example

Differentiate with respect to x:

a) $y = \ln(x\, e^{-x})$ b) $y = \ln\left(\dfrac{x^2}{(x+2)(x-3)}\right)$.

h. Second and higher derivatives

Given a function $f(x)$, the derivative $f'(x)$ is called the **first derivative**.

Definition

The **second derivative** of $f(x)$ is the derivative of $f'(x)$.
We use $f''(x)$, y'', or $\frac{d^2y}{dx^2}$ to denote it:

$$\frac{d^2y}{dx^2} = \frac{d}{dx}\left(\frac{dy}{dx}\right).$$

Example

Find $f''(x)$ given that $f(x) = x^3 - \frac{3}{x}$.

We can continue differentiating to obtain higher-order derivatives.

Definition

The n**th derivative of** y **with respect to** x is obtained by differentiating $y = f(x)$ exactly n times.
We write $f^{(n)}(x)$ or $\frac{d^ny}{dx^n}$.